

1. A light field with amplitude $E_0 = 4 \times 10^9$ V/m propagates from air into a dielectric with $\epsilon_r = 10.9$. Suppose the reflections are completely suppressed via an ideal antireflection coating. What is E_0 inside the dielectric?
2. The electric field (as a spherical wave) is given by

$$E(r, \theta, \phi, t) = A \frac{\sin \theta}{r} \left[\cos(kr - \omega t) - \frac{1}{kr} \sin(kr - \omega t) \right] \hat{\phi}, \text{ where } \omega = kc.$$

- (a) Find the associated magnetic field.
 - (b) Calculate the Poynting vector $\mathbf{S}$. Average $\mathbf{S}$ over a full cycle to get the intensity vector $\mathbf{I}$.
 - (c) Integrate $\mathbf{I} \cdot d\mathbf{a}$ over a spherical surface to determine the total power radiated.
3. Consider a plane wave incident *normally* on a dielectric film of thickness d (see Fig. below). Assume the electric field to be along the y -axis. Use E_{10}^+ and E_{10}^- to represent amplitudes of the forward and backward propagating waves in region 1 and similarly for other regions. The third medium is extended to infinity. k_1, k_2 and k_3 are the propagation vectors in medium 1, 2 and 3 respectively, with $k_i = \frac{\omega}{c} n_i$.

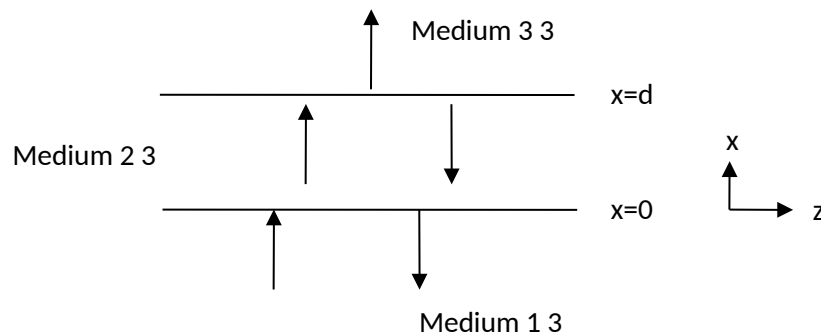


Fig. 1

- (a) Write down the expressions for electric fields in the three media.
 - (b) Write down the corresponding expressions for magnetic fields.
 - (c) Find the amplitude reflection coefficient $r = E_{10}^- / E_{10}^+$ and reflection coefficient R .
 - (d) Show from (c) that, for $\delta = k_2 d = 0$, $R = \left(\frac{n_1 - n_3}{n_1 + n_3} \right)^2$.
4. Calculate the (time-averaged) energy density of an electromagnetic plane wave in a conducting medium (see below equations).

$$E(z, t) = E_0 e^{-kz} \cos(kz - \omega t + \delta_E) \hat{x}$$

$$B(z, t) = B_0 e^{-kz} \cos(kz - \omega t + \delta_E + \phi) \hat{y}$$

Show that the magnetic contribution always dominates.
 5. Show that when a wave undergoes total internal reflection at the interface of two media, the amplitude of the transmitted wave decreases exponentially in the direction perpendicular to its direction of propagation.